



Deep tubewell use is associated with increased household microbial contamination in rural Bangladesh: Results from a prospective cohort study among households in rural Bangladesh[☆]

Varun Goel^{a,*}, Brianna Chan^b, Mia Ziade^c, Md. Yunus^d, Md. Taslim Ali^d, Md. Al Fazal Khan^d, Md. Nurul Alam^d, ASG Faruque^d, Shahabuddin Babu^d, Md. Masnoon Kabir^d, Paul L. Delamater^{a,e}, Marc Serre^b, Mark D. Sobsey^b, Md. Sirajul Islam^d, Michael Emch^{a,b,e}

^a Carolina Population Center, University of North Carolina-Chapel Hill, Chapel Hill, USA

^b Gillings School of Global Public Health, University of North Carolina-Chapel Hill, Chapel Hill, USA

^c Department of Statistics, University of North Carolina-Chapel Hill, Chapel Hill, USA

^d International Centre for Diarrhoeal Disease Research, Bangladesh (icddr), Dhaka, Bangladesh

^e Department of Geography, University of North Carolina-Chapel Hill, Chapel Hill, USA

ARTICLE INFO

Keywords:

WaSH
Bangladesh
Tubewells
Water quality
E. coli
Microbial contamination

ABSTRACT

Deep tubewells are important sources of arsenic mitigation in rural Bangladesh. Compared to commonly available shallow tubewells, deep tubewells tap into deeper low-arsenic aquifers and greatly reduce exposure to arsenic in drinking-water. However, benefits from these more distant and expensive sources may be compromised by higher levels of microbial contamination at point-of-use (POU). This paper examines differences in microbial contamination levels at source and POU among households using deep tubewells and shallow tubewells, and investigates factors associated with POU microbial contamination among deep tubewell users. We assessed a prospective longitudinal cohort of 500 rural households in Matlab, Bangladesh, across 135 villages. Concentration of *Escherichia coli* (*E. coli*) in water samples at source and POU using Compartment Bag Tests (CBTs) was measured across rainy and dry seasons. We employed linear mixed-effect regression models to measure the effect of different factors on log *E. coli* concentrations among deep tubewell users. CBT results show that log *E. coli* concentrations are similar at source and at POU during the first dry and rainy season, but are significantly higher at POU among deep tubewell users during the second dry season. Log *E. coli* at POU among deep tubewell users is positively associated with both presence (exponentiated β $\exp(b) = 2.52$, 95% Confidence Interval (CI) = 1.70, 3.73) and concentration of *E. coli* ($\exp(b) = 1.36$, 95% CI = 1.19, 1.54) at source, and walking time to the tubewell source ($\exp(b) = 1.39$, 95% CI = 1.15, 1.69). Drinking-water during the second dry season is associated with reduced log *E. coli* ($\exp(b) = 0.33$, 95% CI = 0.23, 0.57) compared to the rainy season. These results suggest that while households that use deep tubewells have lower arsenic exposure, they may be at higher risk of consuming microbially contaminated water compared to households that use shallow tubewells.

1. Introduction

Most rural households in Bangladesh obtain their drinking-water from groundwater aquifers through tubewells. The majority of these groundwater sources are shallow tubewells, that generally extract water from the shallow aquifer at depths less than 50 m (Bangladesh Demographic and Health Survey 2017-18, 2020). However, widespread

arsenic contamination in the shallow aquifer resulted in a public health emergency from the exposure of millions of residents to levels of arsenic above permissible limits (Ahmed et al., 2006; Flanagan et al., 2012).

In order to mitigate arsenic exposure, households either switched to neighboring low-arsenic shallow tubewells or to deep tubewells (Van Geen et al., 2002). Switching to low-arsenic shallow tubewells is the most common strategy because shallow tubewells are less expensive and

[☆] This paper has been recommended for acceptance by Sarah Harmon.

* Corresponding author. Carolina Population Center, 123 W Franklin St, Chapel Hill, NC, 27516, USA.

E-mail address: varung@live.unc.edu (V. Goel).

more convenient due to widespread pre-existing infrastructure in rural areas as compared to deep tubewells (Van Geen et al., 2016). However, low-arsenic shallow tubewells are associated with multiple risks such as: 1) poorer water quality due to complex local hydrological factors such as the downward transport of *E. coli* associated with local recharge (Islam et al., 2001; Leber et al., 2011; Van Geen et al., 2011), 2) risk of future arsenic contamination due to groundwater overuse (Mozumder et al., 2020), and 3) increased childhood diarrheal disease (Van Geen et al., 2011; Wu et al., 2011a). On the other hand, deep tubewells tap into aquifers at depths greater than 150 m that are generally free of arsenic (Ravenscroft et al., 2013a,b; Van Geen et al., 2003a, 2007), and are expected to continue to provide arsenic-free water for many decades (Michael and Voss, 2008; Ravenscroft et al., 2013a,b). Additionally, well-constructed deep tubewells are also more likely to be free of fecal contamination compared to shallow tubewells due to increased depths and separation from surface transport of fecal materials (Islam et al., 2001; Luby et al., 2008).

Although deep tubewells are being installed frequently to reduce exposure to arsenic, there are other concerns associated with their potential to provide safe drinking-water. One concern is that because deep tubewells are sparse, inequitably located, and are often not owned by the household using them as a drinking-water source due to higher cost, households may walk further and store their drinking-water for a longer time period (Escamilla et al., 2011; Van Geen et al., 2016; Winston et al., 2013). Such extended water handling and use may increase the risk of microbial contamination at point-of-use (POU), possibly offsetting any additional protection against contamination achieved due to the greater depth from which the water was obtained (Hoque et al., 2006; Howard et al., 2006). Additionally, barriers to deep tubewell access such as distance and ownership may also prompt households to use unsafe water sources for other non-consumption tasks such as washing drinking-water storage containers with surface water (Goel et al., 2020), potentially increasing microbial contamination at POU (Lokuge et al., 2004; Luby et al., 2008).

Current evidence on the association between deep tubewells and microbial contamination at POU is unclear. A previous study by Coccio et al. (2021), found that switching to community deep tubewells, that are often more distant compared to privately owned tubewells, did not increase household exposure to fecal microbial contamination in stored drinking-water containers. Using the first wave of data from our study, we found that over 70% of households using deep tubewells had microbially contaminated drinking-water in their storage containers and deep tubewells located further away exacerbated household microbial contamination risk compared to those located nearby (Goel et al., 2019). Importantly though, while microbial contamination at POU was prevalent, there was no significant difference in microbial contamination between stored drinking-water among households using deep tubewells and those using shallow tubewells. However, many questions remain about the long-term effect of deep tubewell use on microbial contamination risk. For example, there may be important geographical, environmental, social, or economic factors that may modify the association between deep tubewell use and microbial contamination, resulting in heterogeneous impacts, that may otherwise be masked in the previous studies.

In this paper, we build on our previous study (Goel et al., 2019) and address some limitations of previous studies in important ways. First, we compare drinking-water quality among households using deep tubewells compared to shallow tubewells at source and POU across two additional time periods including both the dry and rainy seasons. Second, after controlling for confounding factors and differences among households choosing a deep or a shallow tubewell, we assess the association of deep tubewell use and microbial contamination at POU. Finally, we examine the role of important risk factors, such as tubewell access, source water quality, seasonality, sanitation, and the hygiene environment on microbial water quality at POU among deep tubewell users.

2. Materials and methods

2.1. Study area

The study area is Matlab, Bangladesh, a rural region spanning 184 square kilometers and located 50 km southeast of Dhaka (Alam et al., 2017) in the Chattogram Division (formerly Chittagong Division). It is a field research site for the International Centre for Diarrheal Disease Research, Bangladesh (icddr,b), which has an extensive Health and Demographic Surveillance System (HDSS) covering 142 villages, with a total population of approximately 230,000 individuals and 59,000 households. HDSS collects health information and vital demographic events from each household bimonthly.

Households in Matlab, similar to other parts of rural Bangladesh, are arranged spatially based on patrilineal linkages in clusters called *baris*. A *bari* contains up to a dozen households, with an average of five to six households per *bari*. All members of a *bari* generally drink from their respective own or a shared tubewell source. *Baris* that do not own a tubewell procure drinking-water from tubewells in neighboring *baris* or other communities or facilities, such as a school or religious building. Most individuals living in Matlab obtain water by walking to the source and carrying the water to their household.

2.2. Study design

We conducted a prospective analysis of households with children under five years of age that used either a shallow tubewell or a deep tubewell as their primary source of drinking-water at baseline. Household enrollment occurred from Sept 1, 2016 to Sept 30, 2016. Assuming 90% statistical power and 5% Type 1 error, we estimated a total sample size of 500 households, with 250 households each using shallow tubewells or deep tubewells. Households were selected using a random number generator under a stratified random sampling approach from a sampling frame consisting of all households that reported using a shallow or a deep tubewell based in the Matlab Health and Socioeconomic Survey-2 (MHSS-2) and had at least one child born after Aug 31, 2016 based on the 2016 HDSS. MHSS-2 is a detailed survey of approximately 11% of randomly selected Matlab households for which many household characteristics were collected and we based our sample on a sub-sample of the MHSS-2 sample. Households with under-five children that use a shallow or a deep tubewell as a primary drinking-water source were randomly selected into 250 shallow and 250 deep tubewell households with well depths determined from the MHSS-2 and HDSS wherever applicable. Households were administered a baseline survey on water use, handling, storage, sanitation, and hygiene environment by Community Health Research Workers (CHRWs) during the rainy (June to September) season. They were then followed up three times during the rainy and dry (November to February) seasons to collect water quality samples at tubewell source and POU. Shortly after the third round of follow-up, households were visited by CHRWs to assess general water, sanitation, and hygiene habits during the dry season. The study received approval from the institutional review boards (IRB) of both University of North Carolina and icddr,b, and informed consent was obtained from all participants.

2.3. Exposures and risk factors

CHRWs and field assistants recorded the type of tubewell (i.e., shallow or deep) used by the household for drinking-water based on the depth. Most tubewell owners are aware of the depth of the tubewell based on the length of polyvinyl chloride (PVC) pipe needed to install the tubewell. Owners of the well were asked about the depth. Although the range of depths at which a tubewell is classified as a deep tubewell will vary across different contexts, it is widely accepted that in the context of arsenic mitigation, tubewells at depths greater than 150 m are supposed to be entirely free of arsenic, as opposed to tubewells that have

lower depth (Ravenscroft et al., 2013a,b). In this study, tubewells greater than 150 m were classified as deep tubewells, which is consistent with the classification of deep tubewells used in both previous studies, and by the Department of Public Health and Engineering (DPHE), the government unit responsible for deep tubewell installations (Escamilla et al., 2011; Van Geen et al., 2002, 2003a; 2003b; Winston et al., 2013). It is also important to note that although households were enrolled as deep or shallow tubewell households based on their primary drinking-water source at baseline, some households may switch to another type of tubewell for procuring their drinking-water during a follow-up visit. Hence, the exposure variable (deep or shallow tubewell use) was considered time-varying and was assigned based on the type of tubewell used by the enrolled households at each visit.

Data on water collection, handling, storage, hygiene, and sanitation were collected during both the rainy and the dry season. During collection of microbial data, information on how long drinking-water is stored in the container was collected through a survey by the field team while collecting water samples from the households. Respondents were asked to indicate the approximate time at which they last filled their main drinking-water storage container. The storage time was then calculated as time elapsed between the last refill of the main drinking-water storage container and the time of sample collection. In a few cases, where the main drinking-water container was empty, respondents were asked to provide the time the container was last filled, and a missing value was recorded for POU *E. coli* contamination. The primary tubewell source was also identified, with the respondent walking with the field team to their respective sources. Global Positioning System (GPS) receivers and stopwatches were used to calculate both distance and walking time to the tubewell.

2.4. Outcomes

The field team measured water quality by collecting two water samples from each household: one from the tubewell source, and the other from the household container at POU, during one rainy and two dry seasons across 18 months. These samples were analyzed using the Compartment Bag Test (CBT) for the fecal indicator bacterium *E. coli*. *E. coli* is the fecal indicator bacterium recommended by the World Health Organization (WHO) Guidelines for Drinking-water Quality (GDWQ) (World Health Organization, 2017). The CBT is a microbial water quality field test of simple design consisting of a clear plastic bag with five internal compartments of various volumes to determine a Most Probable Number (MPN) concentration of *E. coli* bacteria per 100 mL. MPN values cover concentrations ranging from an absence of fecal microbial contamination (0 *E. coli* per 100 mL) to extremely high contamination (≥ 100 *E. coli* per 100 mL) (Stauber et al., 2014; Wang et al., 2017). The advantage of MPN quantification is that it gives a range of indicator concentration values across a spectrum of microbial risk ranging from no risk (safe at 0 *E. coli* per 100 mL) to extremely high risk (unsafe at ≥ 100 *E. coli*/100 mL). Hence, the values can be mapped to standardized risk categories provided by the WHO GDWQ (Brown et al., 2008).

2.5. Statistical analysis

We first compared unadjusted differences in source and POU *E. coli* contamination among deep and shallow tubewell users by calculating geometric mean concentrations with 95% confidence intervals of *E. coli* MPN per 100 mL for two dry seasons (survey waves 1 and 3) and one rainy season (survey wave 2). We log-transformed POU *E. coli* concentrations and calculated geometric means to make our estimates less prone to possible skewness due to the upper detection limits of the CBT test and less sensitive to the possibility that results with *E. coli* greater than 100 MPN per 100 mL may have *E. coli* concentrations orders of magnitude greater than those detected. For samples with no detected *E. coli*, we set values to 0.5 MPN per 100 mL to enable log transformation

of water quality data. Then, we analyzed the total effect of tubewell choice for drinking-water on household POU water quality using linear multilevel mixed-effect models. Using the log concentrations of *E. coli* in drinking-water at POU as the dependent variable, we examined changes in the expected geometric mean of POU *E. coli* concentrations based on change in tubewell exposure.

We included covariates in our model based on a conceptual directed acyclic graph (DAG) (Supplementary Figure A1) to account for measured confounding based on the theoretical expectation of relationships between variables and water quality indicators. The minimum set of adjustment covariates include mother's educational attainment, status of tubewell ownership (i.e., owned or not owned by household), and the time taken to walk to the tubewell source to collect drinking-water. In addition, we also added fixed effects for the wave during which the survey was taken and random household effects to account for multiple observations per household across the three survey waves. We also ran unadjusted models (accounting for wave fixed effects and household random effects) for each covariate to explore associations based on the observed data. To measure the effect of different risk factors on water quality among deep tubewell users, we restricted the sample to households that reported using deep tubewells as their primary drinking-water source during the survey wave. These risk factors were selected based on commonly known risk factors affecting POU microbial water quality among households drinking water from tubewells in rural Bangladesh. These factors include water quality at tubewell source, season, time taken to walk to the drinking-water source, ownership of drinking-water source, use of pond water to wash drinking-water storage containers, use of improved latrines, hand-washing with soap, and presence of animal feces in the household compound.

We ran separate multivariate linear mixed effect models for each risk factor and water quality at POU based on conceptual DAGs for each variable. While we were interested in effect sizes specifically for deep tubewell users, we also ran similar models for each risk factor among shallow tubewell users to help place the results and discussion in the larger context of tubewell use in rural Bangladesh. All DAGs were created using Dagitty v3.0 (Textor et al., 2016), and regression analysis was conducted using the package 'lme4' in R v4.0 (Bates et al., 2015; R Core Team, 2020).

3. Results

3.1. Household and tubewell source characteristics

Table 1 displays the important descriptive statistics for the households in the sample. Baseline socioeconomic status (SES) characteristics such as wealth index quintile rank and mother's education were similar for households that use shallow and deep tubewell, with 15% of deep tubewell households in the lowest wealth quintiles and 22% of deep tubewell households in the highest wealth quintiles. Tubewell ownership was higher among shallow tubewell households, with 84–85% of households reporting owning their drinking-water source, compared to 43–44% of deep tubewell households.

3.2. Drinking-water storage container POU characteristics

An aluminum container called a *kolshi* is the most common drinking-water storage container type, with 75% of deep tubewell households reporting using a *kolshi*, compared to 43% of shallow tubewell households. Plastic jugs were the second most common drinking-water storage container type, with 17% of deep tubewell households reporting using a plastic jug, compared to 38% of shallow tubewell households. Deep tubewell households also reported higher median storage volumes (10 L) compared to shallow tubewell households (4 L) and stored water longer both during the rainy and dry seasons. One out four of deep tubewell households reporting drinking-water storage for more than 24

Table 1

Characteristics of households using shallow and deep tubewells as their primary source of drinking-water by season. All values are reported as n (% of households) unless otherwise noted.

Baseline Variables	Shallow (249)		Deep (238)	
Wealth index (%)				
(Poorest) Quantile 1	39 (15)		37 (15)	
(Richest) Quantile 5	56 (22)		54 (22)	
Mother's education in years				
Mean [standard deviation (sd)]	3.91 (3.84)		3.94 (3.96)	
Seasonal Variables	Rainy (247)	Dry (255)	Rainy (238)	Dry (229)
Tubewell ownership				
Own <i>bari</i>	212 (84)	216 (85)	108 (44)	99 (43)
Other <i>bari</i>	36 (14)	32 (13)	107 (44)	86 (38)
Community	5 (2)	7 (2)	31 (13)	44 (19)
Main drinking-water storage container type				
Metal container (<i>kolshi</i>)	109 (43)		185 (75)	
Plastic bucket	16 (6)		7 (3)	
Metal jug	22 (9)		8 (3)	
Plastic jug	96 (38)		41 (17)	
Main drinking-water storage container volume				
[Median liters (inter-quartile range: IQR)]	4 (3–12)		10 (3–13)	
Main drinking-water storage container storage time				
Water is consumed almost immediately	94 (37)	129 (51)	75 (30)	89 (39)
Less than a day (2–24 h)	153 (60)	112 (44)	117 (48)	102 (45)
1 day or more (24–48 h)	6 (2)	14 (5)	54 (22)	38 (17)
Hygiene and Sanitation				
Drinking-water storage container cleaned with surface water	70 (28)	53 (21)	120 (51)	93 (41)
Mother's hands washed with water and soap	179 (72)	191 (75)	139 (58)	196 (86)
Whether animal feces present in the house	116 (47)	126 (49)	109 (46)	80 (35)
Whether household has an improved latrine	211 (85)	210 (86)	193 (81)	207 (90)

h during both rainy and dry seasons.

3.3. Hygiene and sanitation

A higher proportion of deep tubewell users also reported using surface water to wash their drinking-water containers during both the rainy (51%) and dry (41%) seasons, compared to shallow tubewell users. Among mothers from deep tubewell households, handwashing behavior varied seasonally, with 58% reporting handwashing with soap during the rainy season compared to 86% during the dry season. In deep tubewell households 46% had some visible amount of animal feces in the household compound (*bari*) during the rainy season, similar to shallow tubewell households. During the dry season, 35% of households using deep tubewells had some visible amount of animal feces in the compound, compared to 49% in shallow tubewell households. Over 80% of both deep tubewell and shallow tubewell households used an improved latrine, with only one reported case of open defecation in the household cohort.

3.4. Source and POU water quality

Fig. 1A displays the geometric means of *E. coli* MPN for both the tubewell source and the POU drinking-water storage containers. Overall, the geometric mean of *E. coli* MPN per 100 mL was much lower at the tubewell source compared to POU. The geometric means of water quality at the tubewell source were similar among shallow and deep tubewell users and across the three waves. Geometric means of POU

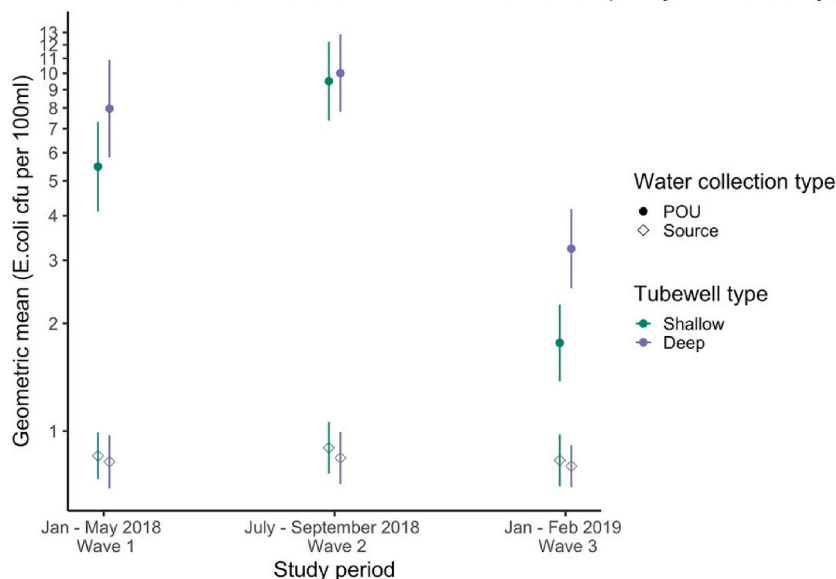
E. coli MPN/100 mL were significantly higher at POU compared to the microbial water quality at tubewell source for both tubewell types. The highest geometric means were observed during the 2018 rainy season (Wave 2: July to September) for both shallow and deep tubewell households, while POU *E. coli* geometric means were observed to be the lowest during the 2019 dry season (Wave 3: January to February). During Wave 3, there was a significant difference in geometric means of *E. coli* MPN concentrations at POU among shallow and deep tubewell users: POU drinking-water storage containers among deep tubewell households had a geometric mean of 3.24 *E. coli* MPN per 100 mL compared to shallow tubewell household containers that had a geometric mean of 1.77 *E. coli* MPN per 100 mL. Fig. 1B characterizes household drinking-water quality based on the standardized WHO risk categories for shallow and deep tubewell households across each wave. During the rainy season (Wave 2), only 16% of deep or shallow tubewell of households did not have detectable *E. coli* concentrations. The proportion of households with no detectable *E. coli* in drinking-water during the dry seasons (Waves 1 and 3) was higher but variable; During Wave 1, 37% of shallow tubewell users had safe drinking-water, compared to 31% of deep tubewell users (χ^2 (Chi-squared test statistic) = 1.84, $p = 0.088$), and during Wave 3, 55% of shallow tubewell users had safe drinking-water, compared to 45% deep tubewell users ($\chi^2 = 4.59$, $p = 0.016$). A higher proportion of households in Wave 1 had drinking-water categorized as low risk as compared to the rainy season (Wave 2) ($\chi^2 = 38.14$, $p < 0.001$). The proportion of households containing drinking-water at very high risk during Wave 1 compared to Wave 2 was marginally higher, but not significant, among both shallow tubewell ($\chi^2 = 0.99$, $p = 0.159$) and deep tubewell users ($\chi^2 = 2.56$, $p = 0.055$).

3.5. Model results

Table 2 displays both the adjusted and unadjusted results from the model evaluating the impact of tubewell type on log *E. coli* MPN per 100 mL at POU. Unadjusted univariate analysis suggests that using a deep tubewell was associated with an expected geometric mean of *E. coli* MPN/100 mL at POU 1.42 (95% CI: 1.13–1.77) times the geometric mean for households drinking-water from shallow tubewells. Factors such as tubewell ownership, time taken to tubewell source, and dry season compared to rainy season were significantly related to POU water quality. Not owning the drinking-water source was associated with an expected geometric mean of POU *E. coli* MPN/100 mL 1.30 (95% CI: 1.03–1.64) times the geometric mean for households that owned their drinking-water source. Similarly, a 1-min increase in time taken to tubewell source was associated with a 29% increase in the POU *E. coli* MPN/100 mL geometric mean (95% CI: 13%–48%), and both dry seasons (Waves 1 and 3) were associated with expected geometric mean of POU *E. coli* MPN/100 mL 0.66 and 0.26 times the expected geometric mean of POU *E. coli* MPN/100 mL during the rainy season (Wave 2). In the adjusted model, the effect of using a deep tubewell is attenuated, but significantly associated with an expected geometric mean of *E. coli* MPN/100 mL at POU 1.30 (95% CI: 1.02–1.67) times the geometric mean for households drinking-water from shallow tubewells.

Adjusted analysis for the effect of each possible exposure on microbial water quality at POU among deep tubewell households suggest that microbial water quality at tubewell source, time to tubewell source, and seasonality are significant predictors of POU water quality (Table 3, Fig. 2). Presence of *E. coli* at the deep tubewell source is associated with an expected geometric mean of 2.52 (95% CI: 1.70–3.73) times compared to deep tubewells without any detectable *E. coli* in drinking-water source. A one-unit increase in log *E. coli* MPN/100 mL at source was associated with a 36% increase (95% CI: 1.19–1.54) in the expected geometric mean of POU *E. coli* MPN/100 mL. The expected geometric mean of POU *E. coli* during the dry season in Wave 3 was 0.33 (95% CI: 0.23–0.47) times the POU *E. coli* values during the rainy season in Wave 2. A 1-min increase in time to the deep tubewell water source was associated with a 39% increase in the expected geometric mean of *E. coli*

A Geometric means of tubewell and POU water quality across study period



B Household microbial contamination risk at POU

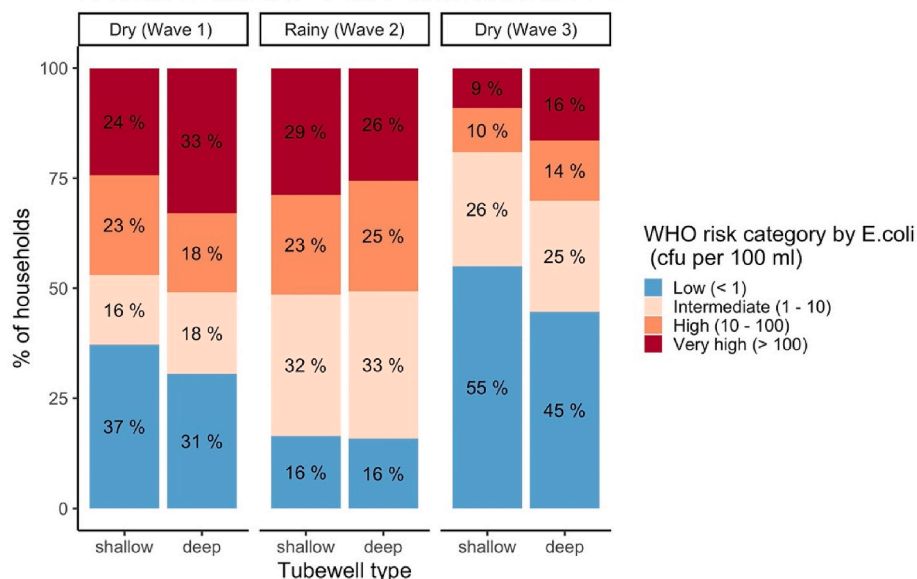


Fig. 1. Microbial contamination at tubewell source and POU. A) Distribution of geometric means (95% CI) of source and POU water quality for shallow and deep tubewell users across study period. B) Household microbial contamination at POU based on risk categories defined by the WHO GDWQ.

at POU (95% CI: 15%–69%). Other factors, such as dry season during Wave 1 compared to the rainy season in Wave 2, ownership of deep tubewell source, washing of drinking-water storage container with surface water, use of an improved latrine, and handwashing with soap among mothers were not significantly associated with change in microbial contamination at POU.

4. Discussion

Deep tubewells are important sources of sustainable arsenic mitigation efforts in rural Bangladesh; however, few studies have demonstrated the real-world effectiveness of these tubewells for providing microbially safe drinking-water to millions of rural Bangladeshis. This study provides further supporting evidence that while deep tubewells have low risk and magnitude of microbial contamination at source, microbial contamination at POU is common and much higher across all time-periods irrespective of tubewell type. In addition, our results also

show that there may be potential for marginally increased microbial contamination at POU among households using deep tubewells compared to those using shallow tubewells, after controlling for confounders. Compared to previous studies examining deep tubewell use and microbial contamination, our study has two major strengths. First, it measures the magnitude of microbial contamination using an affordable field test, as compared to a presence/absence test such as the test used in the [Cocciolo et al. \(2021\)](#) study. Because the microbial contamination at POU is common among both shallow and deep tubewell users, using presence/absence of microbial contamination as an outcome may mask potential differences in the magnitude of exposure to microbial contamination.

Additionally, compared to previous research examining microbial contamination among deep and shallow tubewell users that only investigated one dry season ([Goel et al., 2019](#)), this present study included two more seasons including another dry season. The study reveals the variability in the magnitude of microbial contamination

Table 2

Model results of analysis comparing effect of tubewell type on POU water quality. Exponentiated coefficients and confidence intervals (in parenthesis) from unadjusted and adjusted linear mixed effect models for estimating impact of tubewell type on POU water quality. For categorical variables (tubewell type, ownership, and seasonality), the coefficients represent the ratio of expected geometric means of *E. coli* contamination at POU with a change in variable compared to the reference group. For continuous variables, the coefficients represent the ratio of expected geometric means of *E. coli* contamination at POU with a one unit increase in the predictor variable. Bold values are significant at $p < 0.05$.

Variables	Unadjusted model	Adjusted model
Using a deep tubewell	1.42 (1.13–1.77)	1.30 (1.02–1.67)
Using a shallow tubewell	REF	REF
Not owning the drinking-water source	1.30 (1.03–1.64)	0.95 (0.71–1.28)
Mother's education (no. of years)	1.01 (0.979–1.04)	1.01 (0.98–1.04)
Time to drinking-water source (minutes)	1.29 (1.13–1.48)	1.24 (1.05–1.47)
Wave 1 (Dry season)	0.66 (0.51–0.86)	0.69 (0.53–0.91)
Wave 3 (Dry season)	0.26 (0.20–0.33)	0.25 (0.19–0.33)
Wave 2 (Rainy season)	REF	REF

Table 3

Model results of analysis of risk factor effects on POU water quality among deep tubewell households. Exponentiated coefficients and confidence intervals (in parenthesis) from individual adjusted linear mixed effect models for estimating impact of each variable on POU water quality among deep tubewell users. Bold values are significant at $p < 0.05$

Variable	Beta	CI
Log <i>E. coli</i> at source	1.36	(1.19–1.54)
Presence of <i>E. coli</i> at source	2.52	(1.70–3.73)
Dry season (Wave 1)	0.80	(0.54–1.17)
Dry season (Wave 3)	0.33	(0.23–0.47)
Time to source	1.39	(1.15–1.69)
Not owning the source	0.86	(0.62–1.21)
Washing storage container with surface water	1.02	(0.66–1.58)
Using improved latrine	0.81	(0.57–1.14)
Handwashing with soap	0.94	(0.67–1.33)
Presence of animal feces	1.06	(0.76–1.47)

across time and multiple seasons, and potential small but significant differences in POU microbial quality between households using deep and shallow tubewells during the second dry season. While we do not examine effect modification by season in this analysis, it is possible that during the dry season, risk factors for household contamination linked to longer distance and longer storage times among deep tubewell users may be more important compared to the rainy season, where risk factors linked to higher contamination in a wetter environment in ponds, soil, flies and food may be more pertinent (Ercumen et al., 2017). It should be noted however, that POU microbial contamination levels are often variable (Luby et al., 2015), and since a significant difference in POU microbial contamination among deep and shallow tubewell users was detected during one dry season but not the other, longer follow-up across multiple seasons is needed to determine whether seasonality modifies the relationship between deep tubewell use (as compared to shallow tubewell use) and microbial contamination at POU.

Domestic contamination of drinking-water storage containers is an important pathway of diarrheal diseases and may account for almost one out of every five cases of diarrhea among children under five in parts of rural Bangladesh (Luby et al., 2015). Although they did not test for differences in POU *E. coli* concentrations between tubewell types, Luby et al. (2015) also found that the risk of diarrhea may increase even with modest increases in *E. coli* concentrations. Hence, while POU drinking-water microbial contamination is common in rural Bangladesh irrespective of tubewell type, the potential for additional increased microbial contamination among deep tubewell users may increase the risk of diarrheal diseases compared to the expected risk from shallow tubewells in some contexts. However, it is important to note that

previous studies examining the association between deep tubewells and diarrheal diseases have found conflicting evidence including no significant increase in diarrheal diseases among households using deep tubewells (Wu et al., 2011b), and even a potential reduction in diarrheal disease (Escamilla et al., 2011; Winston et al., 2013). Our study was under-powered to directly examine differences in diarrheal disease incidence based on tubewell type and POU drinking-water microbial quality; Hence, more research incorporating paired frequent and longer microbial testing, tubewell use, and diarrheal disease surveillance for a larger population needs to be conducted to examine whether deep tubewells consistently provide better or worse drinking-water quality at POU and whether it results in higher or lower diarrheal disease incidence compared to shallow tubewells.

Our results also highlight the important risk factors associated with increased risk of microbial contamination at POU among deep tubewell users. The associations between deep tubewell quality at source and household contamination suggest that not only is the presence of *E. coli* at source an important risk factor, but increasing magnitude of *E. coli* at source is also associated with increased levels of contamination in household stored water. Similarly, despite better deep tubewell access in recent years, even a small increase in travel time can have a negative impact on microbial quality in stored household water. While associations between increasing time to tubewell source and household microbial contamination are not surprising in general (Shaheed et al., 2014; World Health Organization, 2017), our results provide additional empirical evidence documenting the relationship between travel time and deep tubewell use in an area of rural Bangladesh. Based on previous literature, it has been suggested that there is a threshold effect where the impact of distance among deep tubewell users on health outcomes such as diarrhea is seen beyond a distance of 150 m (Winston et al., 2013), and that households may be willing to walk up to a distance of 100–150 m to access safe and arsenic-free water (Van Geen et al., 2002, 2003a; 2003b). However, deep tubewell installation and use has significantly increased in rural Bangladesh (Fischer et al., 2020). In our sample, over 80% of deep tubewell users had their drinking-water source within 100 m of their households (Goel et al., 2019), suggesting that previously documented distance thresholds for safe drinking-water decision-making may no longer be relevant.

Our study has several other important policy implications. First, significant differences in both water quality and the factors influencing water quality in household storage for deep and shallow tubewell use suggest that providing safe drinking water to households using deep tubewells would require a separate tailored approach, as opposed to a single intervention for 'tubewell' users. Second, our water sample collection from deep tubewells over a large study area suggests that around 20% of deep tubewells had *E. coli* contamination, similar to levels in shallow tubewells. Because deeper depths should theoretically provide more protection against microbial contamination, it is possible that contamination at deep tubewell source may arise due to poor deep tubewell construction or contamination at the mouth of the handpump used to procure water. It has been shown that handpumps can act as persistent reservoirs for microbial indicator bacteria in rural Bangladesh despite disinfection procedures such as shock chlorination (Ferguson et al., 2011). Additionally, although deep tubewells, especially community deep tubewells, have comparable rates of flushing to shallow tubewells (Wu et al., 2011b), priming with a contaminated source can also provide another avenue for microbial contamination at source (Winston et al., 2013). One option to address such concerns at all stages of water supply, collection, handling, and consumption is through Water Safety Plans (WSPs) of the WHO that address water safety based on comprehensive user guidance (Bartram et al., 2009; Mahmud et al., 2007; World Health Organization (WHO), 2014). While model WSPs have been developed for drinking-water options such as deep tubewells, they have not been updated to reflect the important mechanisms driving risk substitution associated with deep tubewell use. Updated user-friendly WSPs for deep tubewell users that incorporate the

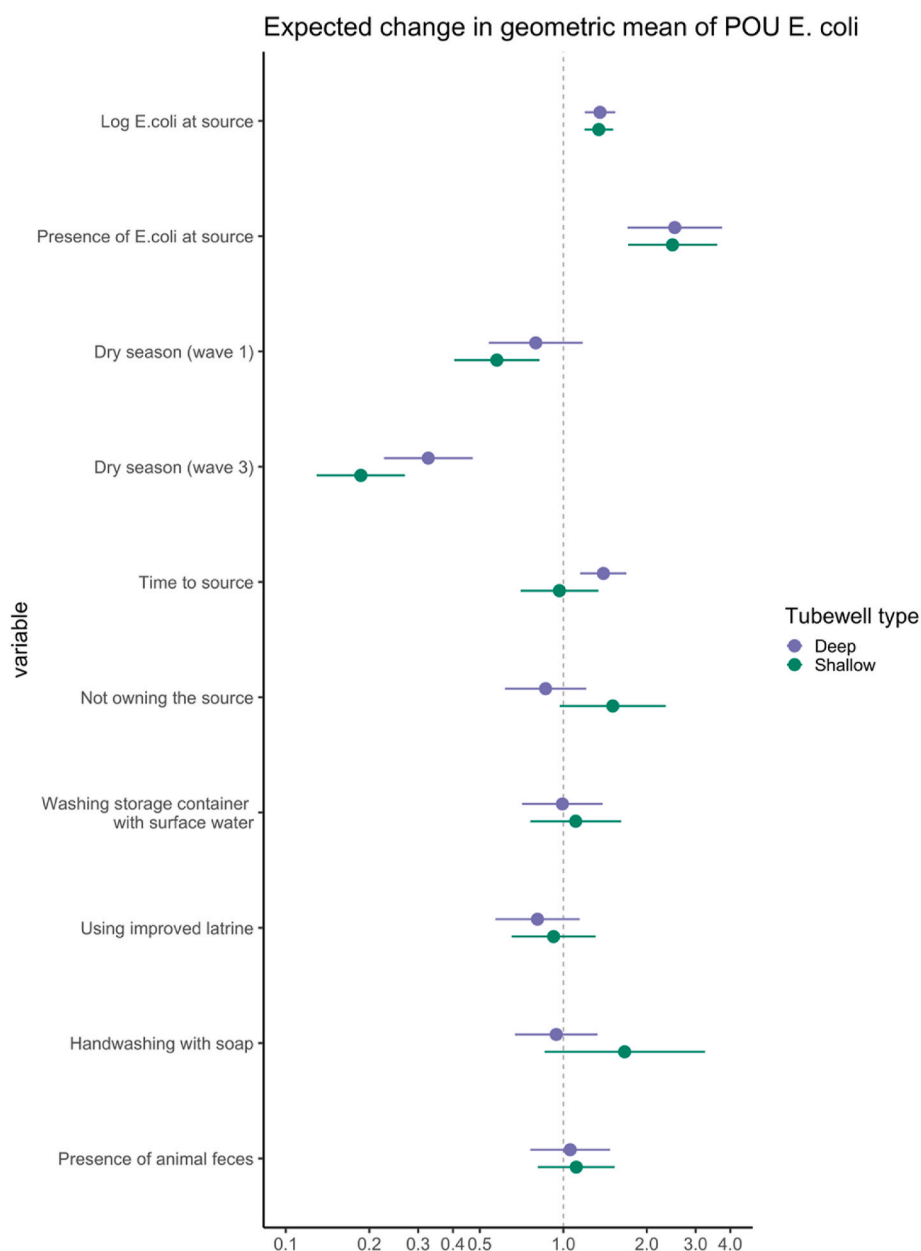


Fig. 2. Comparison of risk factor effects on POU water quality between deep tubewell and shallow tubewell households. Forest plot showing exponentiated coefficients and confidence intervals from individual adjusted linear mixed effect models for estimating impact of each variable on POU water quality among deep tubewell users. Effect sizes for shallow tubewell users are displayed to aid comparison and discussion of results.

potential for risk substitution and consider source contamination as an important risk factor for household drinking-water contamination need to be developed. Additionally, behavioral interventions for safe drinking-water handling and storage, and reduction of unsafe practices such as using surface water for washing storage containers, or priming tubewells with water from contaminated sources should also be implemented to further decrease POU drinking-water microbial contamination among households in rural Bangladesh.

5. Conclusion

Based on the current use and distribution of deep tubewells in Matlab, Bangladesh, we find that, despite their expected benefits of providing chemically and relatively microbially safe drinking-water at source, households that drink water from deep tubewells may potentially be at higher risk of increased microbial contamination in their

stored drinking-water containers compared to households that drink water from shallow tubewells, under certain circumstances. Microbial risk in household drinking-water among deep tubewell users is potentially exacerbated by factors such as increased travel time and microbial contamination at deep tubewell source during the rainy season. As more deep tubewells are installed across rural Bangladesh and other arsenic prone areas to mitigate exposure, policy makers can potentially minimize risk substitution among households through a combination of water quality testing at source, minimizing travel time to deep tubewells, and encouraging better household drinking-water storage practices, especially during the rainy season. In addition to ensuring equitable access to deep tubewells, there is a need for tubewell specific comprehensive user-friendly WSPs, continued behavioral interventions, and community training that address risk substitution and help provide practical guidance for safe water management in communities and households across arsenic affected areas in Bangladesh and beyond.

Credit statement

Varun Goel: Conceptualization, Methodology, Software, Formal analysis, Writing – Original Draft. **Brianna Chan:** Writing – Original Draft, Data Curation. **Mia Ziade:** Software, Formal analysis, Validation, Data Curation, Writing – review & editing. **Md. Yunus:** Funding acquisition, Conceptualization, Writing – review & editing. **Md. Taslim Ali:** Data Curation, Writing – review & editing. **Md. Al Fazal Khan:** Data Curation, Resources, Writing – review & editing. **Md. Nurul Alam:** Data Curation, Writing – review & editing. **ASG Faruque:** Data Curation, Writing – review & editing. **Shahabuddin Babu:** Data Curation, Investigation, Validation, Writing – review & editing. **Md. Masnoon Kabir:** Data Curation, Investigation, Validation, Writing – review & editing. **Paul L. Delamater:** Writing – review & editing. **Marc Serre:** Writing – review & editing. **Mark D. Sobsey:** Funding acquisition, Conceptualization, Writing – review & editing. **Md. Sirajul Islam:** Funding acquisition, Project administration, Conceptualization, Writing – review & editing. **Michael Emch:** Supervision, Funding acquisition, Project administration, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mark Sobsey, a co-investigator and author on this publication is an inventor on the technology being used in this research. Sobsey is also part owner of the company, Aquagenx LLC, which has licensed this technology. UNC-Chapel Hill has the patent on the technology and has a financial interest in the company. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Data availability

The data that has been used is confidential.

Acknowledgements and Funding

This research was funded by the National Science Foundation, grant number BCS-1560970. The research and data collection infrastructure was made possible by the support of icddr,b and its donors who provide unrestricted support to icddr,b for its operation and research. Current core donors providing unrestricted support to icddr,b include the Governments of Bangladesh, Canada, Sweden, and the UK. We also acknowledge the contribution of the whole team of community health workers and the field supervision team at the Matlab icddr,b surveillance site for their data collection efforts and inputs. We are also grateful to the Carolina Population Center for training support and for support by NICHD of the National Institutes of Health under award number P2C HD050924. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2023.121401>.

References

Ahmed, M.F., Ahuja, S., Alauddin, M., Hug, S.J., Lloyd, J.R., Pfaff, A., Pichler, T., Saltikov, C., Stute, M., Geen, A. van, 2006. Ensuring safe drinking water in Bangladesh. *Science* 314, 1687–1688. <https://doi.org/10.1126/science.1133146>.
 Alam, N., Ali, T., Razzaque, A., Rahman, M., Zahirul Haq, M., Saha, S.K., Ahmed, A., Sarder, A.M., Moinuddin Haider, M., Yunus, M., 2017. Health and demographic surveillance system (HDSS) in Matlab, Bangladesh. *Int. J. Epidemiol.* 46, 809–816.

Bangladesh Demographic and Health Survey 2017–18, 2020. . National Institute of Population Research and Training (NIPORT), and ICF, Dhaka, Bangladesh and Rockville, Maryland, USA.
 Bartram, J., Corrales, L., Davison, A., Deere, D., Drury, D., Gordon, B., Howard, G., Rinehold, A., Stevens, M., 2009. Water safety plan manual: step-by-step risk management for drinking-water suppliers, Plans de gestion de la sécurité sanitaire de l'eau : manuel de gestion des risques par étapes à l'intention des distributeurs d'eau de boisson. World Health Organization, Geneva.
 Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Software* 67. <https://doi.org/10.18637/jss.v067.i01>.
 Brown, J.M., Proum, S., Sobsey, M.D., 2008. Escherichia coli in household drinking water and diarrheal disease risk: evidence from Cambodia. *Water Sci. Technol.* 58, 757–763. <https://doi.org/10.2166/wst.2008.439>.
 Coccio, S., Ghisolfi, S., Habib, M.A., Rashid, S.M.A., Tompsett, A., 2021. Do Community Water Sources Provide Safe Drinking Water? Evidence from a Randomized Experiment in Rural Bangladesh. *The World Bank Economic Review*. <https://doi.org/10.1093/wber/lhab006>.
 Ercumen, A., Pickering, A.J., Kwong, L.H., Arnold, B.F., Parvez, S.M., Alam, M., Sen, D., Islam, S., Kullmann, C., Chase, C., 2017. Animal feces contribute to domestic fecal contamination: evidence from E. coli measured in water, hands, food, flies, and soil in Bangladesh. *Environ. Sci. Technol.* 51 (15), 8725.
 Escamilla, V., Wagner, B., Yunus, M., Streafeld, P.K., Van Geen, A., Emch, M., 2011. Effect of deep tube well use on childhood diarrhoea in Bangladesh. *Bull. World Health Organ.* 89, 521–527.
 Ferguson, A.S., Mailloux, B.J., Ahmed, K.M., van Geen, A., McKay, L.D., Culligan, P.J., 2011. Hand-pumps as reservoirs for microbial contamination of well water. *J. Water Health* 9, 708–717.
 Fischer, A., Hope, R., Manandhar, A., Hoque, S., Foster, T., Hakim, A., Islam, MdS., Bradley, D., 2020. Risky responsibilities for rural drinking water institutions: the case of unregulated self-supply in Bangladesh. *Global Environ. Change* 65, 102152. <https://doi.org/10.1016/j.gloenvcha.2020.102152>.
 Flanagan, S.V., Johnston, R.B., Zheng, Y., 2012. Arsenic in tube well water in Bangladesh: health and economic impacts and implications for arsenic mitigation. *Bull. World Health Organ.* 90, 839–846.
 Goel, V., Bell, G.J., Sridhar, S., Islam, M.S., Yunus, M., Ali, M.T., Khan, M.A., Alam, M.N., Faruque, A.S.G., Kabir, M.M., Babu, S., Brandt, K., Shelus, V., Sobsey, M.D., Emch, M., 2020. Considering alternate pathways of drinking-water contamination: evidence of risk substitution from arsenic mitigation programs in rural Bangladesh. *Int. J. Environ. Res. Publ. Health* 17, 5372. <https://doi.org/10.3390/ijerph17155372>.
 Goel, V., Islam, M.S., Yunus, M., Ali, M.T., Khan, A.F., Alam, N., Faruque, A.S.G., Bell, G., Sobsey, M., Emch, M., 2019. Deep tubewell microbial water quality and access in arsenic mitigation programs in rural Bangladesh. *Sci. Total Environ.* 659, 1577–1584. <https://doi.org/10.1016/j.scitotenv.2018.12.341>.
 Hoque, B.A., Hallam, K., Levy, J., Bouis, H., Ali, N., Khan, F., Khanam, S., Kabir, M., Hossain, S., Shah Alam, M., 2006. Rural drinking water at supply and household levels: quality and management. *Int. J. Hyg Environ. Health* 209, 451–460. <https://doi.org/10.1016/j.ijheh.2006.04.008>.
 Howard, G., Ahmed, M.F., Shamsuddin, A.J., Mahmud, S.G., Deere, D., 2006. Risk assessment of arsenic mitigation options in Bangladesh. *J. Health Popul. Nutr.* 24, 346.
 Islam, M.S., Siddika, A., Khan, M.N.H., Goldar, M.M., Sadique, M.A., Kabir, A., Huq, A., Colwell, R.R., 2001. Microbiological analysis of tube-well water in a rural area of Bangladesh. *Appl. Environ. Microbiol.* 67, 3328–3330.
 Leber, J., Rahman, M.M., Ahmed, K.M., Mailloux, B., van Geen, A., 2011. Contrasting influence of geology on E. coli and arsenic in aquifers of Bangladesh. *Groundwater* 49, 111–123.
 Lokuge, K.M., Smith, W., Caldwell, B., Dear, K., Milton, A.H., 2004. The effect of arsenic mitigation interventions on disease burden in Bangladesh. *Environ. Health Perspect.* 112, 1172–1177. <https://doi.org/10.1289/ehp.6866>.
 Luby, S.P., Gupta, S.K., Sheikh, M.A., Johnston, R.B., Ram, P.K., Islam, M.S., 2008. Tubewell water quality and predictors of contamination in three flood-prone areas in Bangladesh. *J. Appl. Microbiol.* 105, 1002–1008.
 Luby, S.P., Halder, A.K., Huda, T.M., Unicomb, L., Islam, M.S., Arnold, B.F., Johnston, R. B., 2015. Microbiological contamination of drinking water associated with subsequent child diarrhea. *Am. J. Trop. Med. Hyg.* 93, 904–911.
 Mahmud, S.G., Shamsuddin, S.A.J., Ahmed, M.F., Davison, A., Deere, D., Howard, G., 2007. Development and implementation of water safety plans for small water supplies in Bangladesh: benefits and lessons learned. *J. Water Health* 5, 585–597. <https://doi.org/10.2166/wh.2007.045>.
 Michael, H.A., Voss, C.I., 2008. Evaluation of the sustainability of deep groundwater as an arsenic-safe resource in the Bengal Basin. *Proc. Natl. Acad. Sci. USA* 105, 8531–8536. <https://doi.org/10.1073/pnas.0710477105>.
 Mozumder, 2020. Origin of groundwater arsenic in a rural pleistocene aquifer in Bangladesh depressurized by distal municipal pumping [WWW Document]. <https://doi.org/10.1029/2020WR027178>.
 R Core Team, 2020. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
 Ravenscroft, P., McArthur, J., Hoque, M., 2013a. Stable groundwater quality in deep aquifers of Southern Bangladesh: the case against sustainable abstraction. *Sci. Total Environ.* 454–455C, 627–638. <https://doi.org/10.1016/j.scitotenv.2013.02.071>.
 Ravenscroft, P., McArthur, J.M., Hoque, M.A., 2013b. Stable groundwater quality in deep aquifers of Southern Bangladesh: the case against sustainable abstraction. *Sci. Total Environ.* 454, 627–638.

- Shaheed, A., Orgill, J., Montgomery, M.A., Jeuland, M.A., Brown, J., 2014. Why “improved” water sources are not always safe. *Bull. World Health Organ.* 92, 283–289. <https://doi.org/10.2471/BLT.13.119594>.
- Stauber, C., Miller, C., Cantrell, B., Kroell, K., 2014. Evaluation of the compartment bag test for the detection of *Escherichia coli* in water. *J. Microbiol. Methods* 99, 66–70. <https://doi.org/10.1016/j.mimet.2014.02.008>.
- Textor, J., van der Zander, B., Gilthorpe, M.S., Liśkiewicz, M., Ellison, G.T., 2016. Robust causal inference using directed acyclic graphs: the R package ‘dagitty’. *Int. J. Epidemiol.* 45, 1887–1894. <https://doi.org/10.1093/ije/dyw341>.
- Van Geen, A., Ahmed, K.M., Ahmed, E.B., Choudhury, I., Mozumder, M.R., Bostick, B.C., Mailloux, B.J., 2016. Inequitable allocation of deep community wells for reducing arsenic exposure in Bangladesh. *J. Water, Sanit. Hyg. Dev.* 6, 142–150. <https://doi.org/10.2166/washdev.2015.115>.
- Van Geen, A., Ahmed, K.M., Akita, Y., Alam, M.J., Culligan, P.J., Emch, M., Escamilla, V., Feighery, J., Ferguson, A.S., Knappett, P., 2011. Fecal contamination of shallow tubewells in Bangladesh inversely related to arsenic. *Environ. Sci. Technol.* 45, 1199–1205.
- Van Geen, A., Ahmed, K.M., Seddique, A.A., Shamsudduha, M., 2003a. Community wells to mitigate the arsenic crisis in Bangladesh. *Bull. World Health Organ.* 81, 632–638.
- Van Geen, A., Ahsan, H., Horneman, A.H., Dhar, R.K., Zheng, Y., Hussain, I., Ahmed, K. M., Gelman, A., Stute, M., Simpson, H.J., 2002. Promotion of well-switching to mitigate the current arsenic crisis in Bangladesh. *Bull. World Health Organ.* 80, 732–737.
- Van Geen, A., Cheng, Z., Jia, Q., Seddique, A.A., Rahman, M.W., Rahman, M.M., Ahmed, K.M., 2007. Monitoring 51 community wells in Araihaazar, Bangladesh, for up to 5 years: implications for arsenic mitigation. *J. Environ. Sci. Health Part A* 42, 1729–1740.
- Van Geen, A., Zheng, Y., Versteeg, R., Stute, M., Horneman, A., Dhar, R., Steckler, M., Gelman, A., Small, C., Ahsan, H., 2003b. Spatial variability of arsenic in 6000 tube wells in a 25 km² area of Bangladesh. *Water Resour. Res.* 39.
- Wang, A., McMahan, L., Rutstein, S., Stauber, C., Reyes, J., Sobsey, M.D., 2017. Household microbial water quality testing in a Peruvian demographic and health survey: evaluation of the compartment bag test for *Escherichia coli*. *Am. J. Trop. Med. Hyg.* 96, 970–975. <https://doi.org/10.4269/ajtmh.15-0717>.
- Winston, J.J., Escamilla, V., Perez-Heydrich, C., Carrel, M., Yunus, M., Streatfield, P.K., Emch, M., 2013. Protective benefits of deep tube wells against childhood diarrhea in Matlab, Bangladesh. *Am. J. Publ. Health* 103, 1287–1291.
- World Health Organization, 2017. Guidelines for drinking-water quality, 4th edition, incorporating the 1st addendum [WWW Document]. URL <https://www.who.int/publications/i/item/9789241549950>. accessed 11.28.20.
- World Health Organization (WHO), 2014. Water Safety Plan (WSP): a Risk Based Approach for Water Safety.
- Wu, J., van Geen, A., Ahmed, K.M., Alam, Y.A.J., Culligan, P.J., Escamilla, V., Feighery, J., Ferguson, A.S., Knappett, P., Mailloux, B.J., McKay, L.D., Serre, M.L., Streatfield, P.K., Yunus, M., Emch, M., 2011a. Increase in diarrheal disease associated with arsenic mitigation in Bangladesh. *PLoS One* 6 (12), e29593. <https://doi.org/10.1371/journal.pone.0029593>.
- Wu, J., Yunus, M., Streatfield, P.K., van Geen, A., Escamilla, V., Akita, Y., Serre, M., Emch, M., 2011b. Impact of tubewell access and tubewell depth on childhood diarrhea in Matlab, Bangladesh. *Environ. Health* 10 (1), 109.